

Foundational Research & Advances in Hidden Subgroup Problem (HSP)

Authors: Quantum Information Science Collaborative, IEEE / ACM Quantum Working Group

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Hidden Subgroup Problem (HSP) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"All the most famous quantum speedups are actually the same puzzle in disguise: finding a hidden secret pattern inside an algebraic symmetry group."

3. Mathematical Formulation & Unitary Rule

$$f(g_1) = f(g_2) \iff g_1 H = g_2 H, \quad |G\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle$$

Quantum Fourier Transform over group representations projects state directly into subgroup annihilators.

4. Step-by-Step Circuit Sequence

Uniform superposition over group elements $|G\rangle$, oracle U_f , measure second register, QFT over group G on first register.

5. Executable IBM Qiskit (Python) Code

```
# Standard coset state representation in Abelian HSP
# |psi_coset> = 1/sqrt(|H|) * sum_{h in H} |g + h>
# QFT_G transforms coset states into subgroup characters orthogonal to H
print("Abelian HSP is efficiently solvable in poly(log |G|) on quantum processors.")
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

$\exp(\Omega(\sqrt{\log |G|}))$ for general G vs $\text{poly}(\log |G|)$ for all finitely generated Abelian groups

7. Key Applications & Future Research Directions

- Unifying foundation for integer factoring and discrete logarithms
- Theoretical basis for exploring Graph Isomorphism quantum algorithms
- Quantum cryptanalysis of group-theoretic cryptographic schemes

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Algorithms pipelines.